PUSHFORWARDS COMMUTE WITH BASE CHANGE ALONG CARTESIAN SQUARES OF BIG AND SMALL ETALE SITES

THE STACKS PROJECT AUTHORS

Editorial note. The source offers 'can be proved in the same manner or can be deduced from Sites, Lemma localize-morphism' for the second equality; the first-equality computation is complete in-span and the second is the identical parallel computation, carried as an explicit core obligation.

Allowed standard prerequisites (editorial).

- Pushforward formulas. lemma-morphism-big-small-etale and lemma-morphism-big-etale (same chapter; former carded L26666, latter NOT carded, declared external). The rules F(U/X) used to evaluate both composites.
- Localize-morphism base change. Sites, lemma localize-morphism (cross-chapter, named). The very-general alternative route for the second identity.

Source reference. Original lemma 0DF5: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 608–608. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DF2: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 511–511. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03CF: focused-seeds/geometry/S9-topo-morphism-big-small-cartesian-diagram-etale/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Consider a cartesian diagram*

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X \end{array}$$

in $(Spaces/S)_{étale}$. Then $i_g^{-1} \circ f_{big,*} = f'_{small,*} \circ (i_{g'})^{-1}$ and $g_{big}^{-1} \circ f_{big,*} = f'_{big,*} \circ (g'_{big})^{-1}$.

Date: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Since the diagram is cartesian, we have for U'/X' that $U' \times_{X'} Y' = U' \times_X Y$. Hence both $i_g^{-1} \circ f_{big,*}$ and $f'_{small,*} \circ (i_{g'})^{-1}$ send a sheaf $\mathcal{F}$ on $(\mathit{Spaces}/Y)_{\acute{e}tale}$ to the sheaf $U' \mapsto \mathcal{F}(U' \times_{X'} Y')$ on $X'_{\acute{e}tale}$ (use Lemmas 0DF2 and 0DF5). The second equality can be proved in the same manner or can be deduced from the very general Sites, Lemma 03CF. $\square$